Anant Parashar

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Work Experience

Software Engineer — AI Applications Apr 2026 – Present
Intoglo *Delhi, India*

- Built end-to-end **AI-powered logistics SaaS** (3 services, ~160K LOC) where every core workflow is GenAI-driven — document classification, data extraction, copilot assistance, and self-improving **ML pipelines** across 17 shipping carriers.
- Designed a **self-learning end-to-end ML pipeline** with 40+ extractors: reverse-engineers regex patterns from PDF text, persists LLM-generated strategies in a **MongoDB knowledge base** with per-field accuracy tracking, and auto-consolidates proven patterns into a **generic fallback** — new carriers get day-one accuracy with **zero config**. Achieved **97% on invoices**, **100% on bills of lading**.
- Built a **cost-optimized 4-tier classification pipeline**: template-hash cache → rule-based scorers → Gemini LLM (only on ambiguous docs) → OCR re-classification for scanned PDFs — **reduced LLM inference cost by ~80%** vs. LLM-first approach.
- Implemented an **LLM strategy builder** that generates extraction pipeline configs from raw PDF text, with a **3-phase automated model evaluation and deployment framework** (shadow → confidence → independent) — an **AI agent that writes and validates its own extraction logic**.
- Designed an **AI copilot** (AWS Bedrock) with **agent orchestration and structured tool use** — LLM invokes typed tools (shipment lookup, SLA check, workflow query) with session persistence and model routing.
- Built **inbound email intelligence pipeline**: DB-driven routing → AI classification → entity extraction → shipment matching by composite key, with a **canonical field authority engine** resolving conflicts across 3 data sources via priority rules and provenance audit trail.
- Built a **22-step workflow state machine** with conditional transitions, parallel cascading, approval gates, least-loaded auto-assignment, and **RBAC across 6 roles** with desk-level scoping — the orchestration layer AI extraction feeds into.

Software Engineer — Backend Systems Dec 2025 – Apr 2026
Global ITCombine (Client: OneAssist / ITC)

- Reduced API latency by **35%** across services handling **10K+ daily requests** through query plan analysis, caching layers, and data access pattern redesign.
- Built **async REST APIs** (FastAPI) and resolved **high-concurrency bottlenecks** in production through SQL optimization and schema index redesign.

Software Engineer — Backend Systems Aug 2023 – Dec 2025
IDEMIA

- Built **secure enterprise backend services** for high-volume identity verification; improved system reliability by **25%** through multi-service workflow debugging and race condition resolution.
- Optimized database performance on large-scale datasets via **index tuning, query plan analysis, and schema normalization**.

Projects

Self-Learning Document Intelligence Engine
Python, FastAPI, Gemini API, TensorFlow, RapidOCR, MongoDB

- Built a **config-driven extraction engine** with 30+ composable pipeline operators — new carrier support requires **zero code changes**, only a JSON config. Includes **"Correct & Train" feedback loop** that auto-learns patterns from human corrections.
- Built **OCR fallback pipeline** (RapidOCR/ONNX + OpenCV + spatial reconstruction) for scanned and corrupted-font PDFs. Deployed on **Kubernetes** with S3 storage and MongoDB persistence.

AI Copilot with Agent Orchestration
AWS Bedrock, LangChain, TypeScript, MongoDB

- Built an **agentic AI assistant** where the LLM decides which tools to invoke based on natural language queries, with **session persistence**, model routing, and **structured output parsing** enabling downstream automation.

Technical Skills

AI / GenAI: LLM Application Design, Prompt Engineering, Model Evaluation, Fine-Tuning, Agent Orchestration, Multimodal AI, LangChain, TensorFlow, Gemini API, AWS Bedrock, Structured Tool Use, ML Pipelines, RAG, NLP, OCR (RapidOCR/ONNX)
Languages: Python, TypeScript, JavaScript, SQL, Java
Backend: Node.js, Express, FastAPI, Microservices, Distributed Systems, AsyncIO
Frontend: Next.js, React 19, Tailwind CSS, NextAuth, Firebase
Databases: MySQL, MongoDB, Redis, Query Optimization, Schema Design
Cloud & Infra: GCP (Vertex AI), AWS (S3, Bedrock, EC2), Docker, Kubernetes (Helm), CI/CD
Core CS: DSA, System Design, Multi-Tenancy, RBAC, Concurrency, Cost-Aware Model Routing

Education

Bachelor of Technology (B.Tech) — Electronics and Communication Engineering Jul 2019 – Jun 2023
JSS Academy of Technical Education, Noida *CGPA: 8.02/10*